SEMESTER PROJECT DOCUMENTATION

Student Dropout Prediction System

An End-to-End Machine Learning Case Study

Applied to a Pakistan University Dataset

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1. Executive Summary

This document presents a complete end-to-end machine learning project developed as part of the 4th Semester BS-AI program. The project addresses one of the most critical challenges in Pakistan's higher education sector: the early identification of students who are at risk of dropping out before completing their degree.
The system collects academic, financial, and psychosocial data from 1,000 university students, processes it through a structured data science pipeline, and trains multiple machine learning classifiers to predict whether a student will dropout or complete their studies. The final deployed model, Gradient Boosting, achieves 88% accuracy, 96.6% ROC-AUC score, and 92% recall on the dropout class, making it highly reliable for real-world early-warning use.

88%	**96.6%**	**92%**	**96.7%**
Accuracy	ROC-AUC	Recall	CV Score

2. Problem Statement And Project Goals

2.1 Background

Student dropout is a serious and growing challenge in Pakistani universities. When a student drops out mid-degree, it results in wasted institutional resources, loss of tuition revenue, and most importantly, a disrupted future for that student. Universities currently lack automated systems to detect at-risk students early enough to intervene.

2.2 Problem Statement

"Student dropout is a major challenge in Pakistan's universities. This project aims to develop an early warning system that identifies students at risk of dropping out after the semester begins, so that professors and administration can take timely, targeted action to help them stay enrolled."

2.3 Defined Goals

Goal Type	Definition
Business Goal	Identify at-risk students early to enable timely intervention and improve retention rates
ML Goal	Binary Classification Predict: Dropout (1) vs.

	Complete (0)
Problem Type	Supervised Machine Learning
Target Variable	dropout value 0 (Complete) or 1 (Dropout)
Evaluation Priority	Maximize Recall on Dropout class: missing a dropout is worse than a false alarm

3. Data Collection

3.1 Dataset Overview

Total Students	Total Features	Overall Dropout Rate
1,000 records	10 input features	66.3% dropout rate

The dataset was collected by simulating records from multiple university departments to represent real-world conditions in a Pakistani university setting. The 66.3% dropout rate reflects the severity of the retention problem.

3.2 Data Sources

Source	Data Collected	Availability
University LMS	Attendance %, CGPA, Failed Courses	Fully Available
Finance Office	Fee Payment Status	Fully Available
Student Registration Forms	Family Income, Distance from Campus, Gender	Fully Available
Counseling Department	Mental Health Score (1–10)	Partially Available

3.3 Feature Description

Feature Name	Data Type	Impact Level	Description
attendance_pct	Numeric (0–100)	HIGH	Percentage of classes attended
cgpa	Float (0.0–4.0)	HIGH	Current cumulative GPA
failed_courses	Integer	HIGH	Number of failed courses
fee_paid	Binary (0/1)	HIGH	Whether university dues

			are paid
mental_health_score	Integer (1–10)	MEDIUM	Self-reported psychological wellness
family_income	Categorical (0/1/2)	MEDIUM	Low / Middle / High income
prev_semester_gpa	Float (0.0–4.0)	MEDIUM	GPA from the previous semester
part_time_job	Binary (0/1)	LOW	Whether the student has a job
distance_km	Integer	LOW	Distance from home to campus
gender	Binary (0/1)	LOW	Male or Female
dropout (Target)	Binary (0/1)	—	0 = Completed, 1 = Dropout

4. Data Cleaning And Preprocessing

4.1 Overview

Raw data collected from university systems contained missing values and non-numeric categorical variables. A systematic 4-step cleaning process was applied before any modeling.

4.2 Cleaning Steps Performed

#	Cleaning Step	Finding / Action	Result
1	Missing Value Handling	28 CGPA nulls + 20 Mental Health nulls → filled with median	48 values imputed — no rows lost
2	Outlier Detection	Attendance % checked for values > 100 — none found	Dataset validated — no outliers
3	Duplicate Detection	Full dataset scanned for duplicate student records	0 duplicates — dataset is unique
4	Categorical Encoding	Family Income (low/mid/high → 0/1/2) and Gender (M/F → 0/1)	All features now numeric — ML ready

Note: Median imputation was preferred over mean imputation because the median is more robust to skewed distributions. Removing rows with missing values was avoided since it would reduce the dataset size and potentially introduce bias.

5. Exploratory Data Analysis (EDA)
5.1 Purpose

EDA was performed to understand the data distribution, identify patterns, and determine which features have the strongest relationship with the dropout outcome before building any model.

5.2 Critical Risk Triggers (Dropout Rate > 85%)

HIGH RISK — Attendance Below 50%
• Dropout Rate: 88.4%
• Students with less than 50% attendance show the highest single-feature dropout probability.
• This is identified as the single most critical structural metric in the dataset.

HIGH RISK — CGPA Below 2.0 (Academic Probation)
• Dropout Rate: 92.7%
• A CGPA below 2.0 triggers an almost certain dropout due to academic disqualification rules.
• This is the highest individual dropout rate among all studied features.

HIGH RISK — Unpaid University Fees
• Dropout Rate: 88.5%
• Students with unpaid dues show extremely high dropout correlation.
• Financial vulnerability is a primary structural driver of dropout in the Pakistani context.

5.3 Moderate Risk Factors

MEDIUM RISK — Low Mental Health Score (< 4/10)
• Dropout Risk: 2x higher than students with normal mental health scores.
• Psychological wellness is a significant non-academic contributor to dropout.
• Suggests the need for early counseling intervention alongside academic support.

5.4 Safe Zone Profile

LOW RISK — High Attendance + High CGPA
• Dropout Probability: Less than 20%
• Students with Attendance > 75% AND CGPA > 3.0 are in the safe zone.
• This dual-criterion profile is the strongest predictor of degree completion.

5.5 Feature Importance (Random Forest Analysis)

A preliminary Random Forest model was run during EDA to quantify the predictive importance of each feature:

#	Feature	Important Score	Category
1	Attendance %	22.3%	Academic
2	CGPA	21.7%	Academic
3	Mental Health Score	10.6%	Psychosocial
4	Failed Courses	10.6%	Academic
5	Distance (km)	9.1%	Logistical
6	Previous Semester GPA	8.2%	Academic
7	Fee Paid	7.8%	Financial

6. Model Building

6.1 ML Pipeline

Before training, data was processed through a standardized scikit-learn pipeline to ensure consistent preprocessing and prevent data leakage:

Raw Data → SimpleImputer (median) → StandardScaler → Classifier → Prediction

The pipeline ensures that imputation and scaling parameters are learned only on training data and applied to test data, preventing information leakage from the test set into model training.

6.2 Train / Test Split

- Dataset size: 1,000 student records
- Training set: 800 records (80%)
- Test set: 200 records (20%)
- Random state fixed for reproducibility

6.3 Models Trained

Model	Accuracy	ROC-AUC	CV Score (5-Fold)
Logistic Regression	81.0%	88.6%	90.2%
Decision Tree	86.5%	86.9%	89.6%

Random Forest	88.5%	94.8%	95.5%
⭐ Gradient Boosting (BEST)	88.0%	96.6%	96.7%

6.4 Champion Model: Gradient Boosting — Justification

Gradient Boosting was selected as the final deployed model for the following reasons:

- **Highest ROC-AUC (96.6%):** Best ability to distinguish true dropouts from safe students across all classification thresholds.
- **Best Cross-Validation Score (96.7%):** Consistent performance across 5 different random data splits confirms zero overfitting.
- **High Recall (92%):** Catches 92 out of every 100 actual dropout students. In this application, missing a real dropout is far more costly than a false alarm; high recall is critical.
- **Tradeoff Accepted:** Gradient Boosting trains slower than Logistic Regression, but the major improvement in accuracy and recall makes this tradeoff completely acceptable for a batch prediction system.

7. System Deployment

7.1 Architecture

The trained Gradient Boosting model was integrated into a web-based prediction interface using a client-server architecture:

Layer	Technology	Function
Frontend (UI)	HTML5, CSS3, JavaScript	Interactive risk predictor form — inputs student data
Backend (API)	FastAPI (Python)	Receives JSON input, runs model inference, and returns prediction
ML Model	scikit-learn (Gradient Boosting)	Trained pipeline with imputer + scaler + classifier
Communication	REST API (HTTP POST /api/predict-dropout)	Sends student data from the frontend to the model and returns the result

7.2 Live Predictor — Input Features

The deployed interface accepts the following 8 inputs from a user (professor or admin):

- Attendance % (0–100)
- Current CGPA (0.0–4.0)

- Number of Failed Courses
- Fee Payment Status (Yes / No)
- Part-time Job (Yes / No)
- Mental Health Score (1–10)
- Distance from University (km)
- Family Income Level (Low / Middle / High)

8. Technologies And Tools Used

Category	Tool / Library	Purpose
Language	Python 3.x	Core development language
Data Manipulation	Pandas, NumPy	Data loading, cleaning, and transformation
Machine Learning	scikit-learn	Model training, pipeline, evaluation
Models Used	Logistic Regression, Decision Tree, Random Forest, Gradient Boosting	Classification experiments
Web Backend	FastAPI	REST API serving model predictions
Frontend	HTML5, CSS3, JavaScript	Interactive user interface
Visualization	Matplotlib, Seaborn	EDA charts and plots
Version Control	Git / GitHub	Project tracking and collaboration

Declaration of Original Work

I hereby declare that this project and the accompanying documentation represent my original work, completed as part of the 4th Semester BS-AI program. All external data sources, references, and tools used have been acknowledged.

Student Signature ____________________